

18. Isotopes of an element differ in -

- (a) the number of protons
- (b) the mass number
- (c) the number of electrons
- (d) the atomic number
- (e) None of the above / More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)

Elements with same atomic number are called isotopes, i.e. elements having same atomic number (number of protons) but different mass number or atomic mass are called isotopes. And the elements with the same atomic mass are called isobars.

19. Which of the following is an incorrect option?

- (a) The atom is electrically neutral due to the presence of protons and electrons inside the nucleus.
- (b) Electrons are negatively charged and have negligible mass.
- (c) Atomic number is the number of protons in an atom of an element.
- (d) Isotopes are useful as nuclear fuel, in medical field, in carbon dating etc.
- (e) None of the above / More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (a)

In the given statements, option (a) is not correct, because the atom is electrically neutral due to the same number of protons (positively charged particles) and electrons (negatively charged particles) in the atom, but only protons and neutrons are lie in the nucleus of the atom and electrons lie outside the nucleus. Statements of options (b), (c) and (d) are correct.

Physical and Chemical Changes, Solution

1. Chemotherapy is

- (a) use of heat for therapeutic effect
- (b) use of chemical for therapeutic effect
- (c) use of natural reagent for therapeutic effect
- (d) All of the above
- (e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (b)

Chemotherapy is a drug treatment that uses powerful chemicals to kill fast-growing cells in your body. Chemotherapy is most often used to treat cancer, since cancer cells grow and multiply much more quickly than other cells in the body. Many different chemotherapy drugs are available. Chemotherapy drugs can be used alone or in combination to treat a wide variety of cancers. Though chemotherapy is an effective way to treat many types of cancer, chemotherapy treatment also carries a risk of side effects. Some chemotherapy side effects are mild and treatable, while others can cause serious complications.

2. An example of physical change –

- (a) Blackening of silverware
- (b) Burning of Candle
- (c) Making of curd from milk
- (d) Dissolving of sugar in water

38th B.P.S.C. (Pre) 1992

Ans. (d)

Physical changes affect the form of a chemical substance, but not its chemical composition. This fact contrasts with the concept of chemical change in which the chemical composition of the substance changes or one or more substances combines or break up to form new substances. In general, a physical change is using physical means. For example, sugar dissolved in water can be recovered by allowing the water to evaporate.

3. Pasteurization is a process by which –

- (a) Milk is preserved at a very low temperature for 24 hours
- (b) Milk is heated for 8 hours
- (c) First milk is heated up to a longer time and then cooled suddenly within specified time
- (d) None of these

38th B.P.S.C. (Pre) 1992

Ans. (c)

In order to pasteurize milk, first, the milk is heated up for longer time and then cooled suddenly within specified time. Pasteurization is a process invented by French scientist Louis Pasteur during the nineteenth century. In 1864, Louis Pasteur discovered that heating beer and wine was enough to kill most of the bacteria that caused spoilage, and hence prevented these beverages from turning sour. This was achieved by eliminating pathogenic microbes and lowering microbial numbers to prolong the quality of the beverage.

4. Milk is an example of :

- (a) A gel (b) A foam
(c) An emulsion (d) A suspension

42nd B.P.S.C. (Pre) 1997

Ans. (c)

An emulsion is a colloid of two or more immiscible liquids where one liquid dispersed in the other. Milk is an example of an emulsion.

5. Which of the following is not a common example of colloid?

- (a) Soda water (b) Milk
(c) Milk of magnesia (d) Fog

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (a)

Collision theory basically explains how reactions occur and why different reactions have different reactions rate. Colloids are classified according to the state of the dispersed phase and the dispersion medium. The common example of colloid are milk, milk of magnesia, fog etc. Whereas soda water is an example of colloidal solution.

6. Colloids have particle size range between

- (a) 10^{-7} m to 10^{-10} m (b) 10^{-7} m to 10^{-10} m
(c) 10^{-10} m to 10^{-12} m (d) 10^{-5} m to 10^{-7} m
(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (a)

Colloids have the particle size range between 10^{-7} m to 10^{-10} m. A colloid is a mixture in which one substance consisting of microscopically dispersed insoluble particles is suspended throughout another substance.

7. The technique of chromatography is used to –

- (a) Identify colour substances
(b) Determine the structure of substances
(c) Dry distillation of colouring substances
(d) Separate the substances from a mixture

42nd B.P.S.C. (Pre) 1997

Ans. (d)

Chromatography is a separation technique used to separate the different components in a liquid mixture. The separation is based on differential partitioning between the mobile and stationary phases. Chromatography may be preparative or analytical. The purpose of preparative chromatography is to separate the components of a mixture for more advanced use and thus it is a form of purification.

8. Which of the following is not a mixture?

- (a) Glass (b) Steel
(c) Aluminium (d) Graphite

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (*)

Glass and steel are mixture but graphite and aluminium are not mixture, graphite is an allotrope because it is a pure element which cannot split into its constituent particles. Whereas glass is a homogeneous mixture of super-cooled molten sodium silicate and calcium silicate, steel is an alloy of iron and carbon with improved strength and fracture resistance compared to other form of iron and aluminium is itself an element (metal). Hence both option (c) and (d) are correct answer.

9. Substances that readily absorb moisture from other substances without chemically reacting with them are called

- (a) desiccants
(b) deliquescent substances
(c) efflorescent substances
(d) dehydrating agents
(e) None of the above/More than one of the above

B.P.S.C. CDPO 2022

Ans. (a)

Desiccating agents readily absorb moisture from other substances without chemically reacting with them. They are also called drying agents. For example - Anhydrous calcium chloride (CaCl_2) and anhydrous zinc chloride (ZnCl_2) are used to absorb moisture from the air.

10. Process of solid camphor into camphor vapour is called:

- (a) Vaporisation (b) Freezing
(c) Melting (d) Sublimation

44th B.P.S.C. (Pre) 2000

Ans. (d)

The sublimation is the process in which some substances upon heating directly changes into vapour form and when cooled again directly turns into solid form. e.g. Solid camphor directly changes into camphor vapour

11. The conversion of a solid directly into gas is called as :

- (a) Sublimation (b) Condensation
(c) Evaporation (d) Boiling
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (a)

See the explanation of the above question.